MA388 Sabermetrics: Lesson 30

Equivalence Coefficient

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Don Mattingly ("Donnie Baseball") - Should he be in the Hall of Fame?

Excerpts from *Here's the HOF case for and against Mattingly* by Paul Casella on mlb.com.

> "Mattingly was arguably the best player in Major League Baseball for a six-year span from 1984-89. He averaged 27 homers and 114 RBIs, while putting up a .327/.372/.530 batting line in those six campaigns. The .530 slugging percentage led all qualified hitters, while Mattingly also paced the Majors in extra-base hits (428) and RBIs (684) during that six-year span."

> "Though he hit a respectable 222 homers, Mattingly wasn't just a power-hitting first baseman. He finished his career with a .307 average, making him one of just eight first basemen in Major League history to hit at least .305 with 200 homers. That list includes five Hall of Famers in Lou Gehrig, Jimmie Foxx, Johnny Mize, Hank Greenberg, and Jim Bottomley."

> "Though Mattingly was one of the game's elite talents from 1984-89, he was an entirely different player starting with the '90 campaign in which he was limited to just 102 games due to a congenital disk deformity in his back. Mattingly averaged only 10 homers and 64 RBIs over his final six seasons, while hitting .286/.345/.405. He topped out at 17 homers and 86 RBIs during that stretch. Though his superb defense continued throughout his career, Mattingly ended his career by hitting .288 with seven homers and 49 RBIs over 128 games in '95."

Career Statistics

Let's look at Don Mattingly's career batting statistics.

```
library(Lahman)
library(tidyverse)
library(knitr)

vars <- c("G", "AB","R", "H", "X2B","X3B", "HR", "RBI", "BB", "SO", "SB",
          "SH", "HBP", "SF")
Batting %>%
  filter(playerID == "mattido01") %>%
  summarise_at(vars, sum) %>%
  mutate(AVG = round(H / AB, 3)) %>%
  select(G, AB, H, HR, AVG) %>%
  kable(caption = "Mattingly Career Statistics", digits = 3)
```

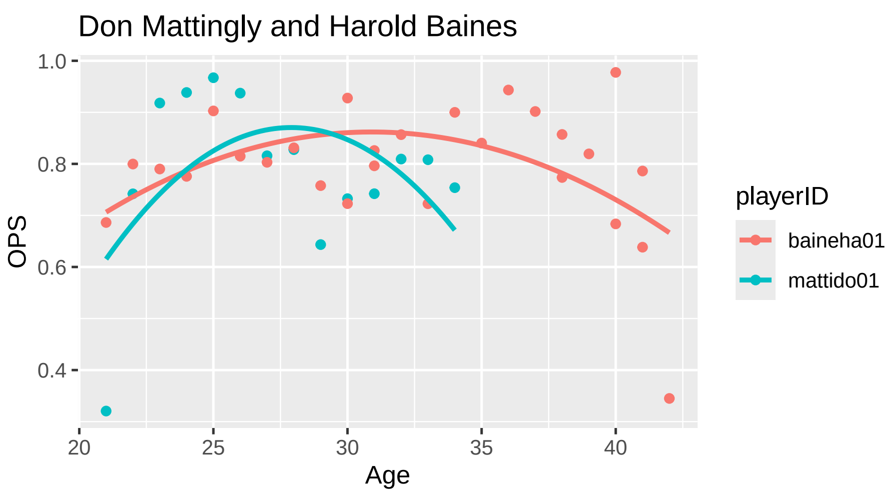
Table 1: Mattingly Career Statistics

G	AB	H	HR	AVG
1785	7003	2153	222	0.307

A picture is worth a thousand words.

```
get_stats <- function(player){
  Batting %>%
    filter(playerID == player) %>%
    inner_join(People, by = "playerID") %>%
    mutate(birthyear = ifelse(birthMonth >= 7,
                              birthYear + 1, birthYear),
           Age = yearID - birthyear,
           SLG = (H - X2B - X3B - HR +
                    2 * X2B + 3 * X3B + 4 * HR) / AB,
           OBP = (H + BB + HBP) / (AB + BB + HBP + SF),
           OPS = SLG + OBP) %>%
    select(playerID, Age, SLG, OBP, OPS)
}

stats <- c("mattido01","baineha01") %>% map_df(get_stats)

stats %>%
  ggplot(aes(x = Age, y = OPS, color = playerID)) +
  geom_point() +
```

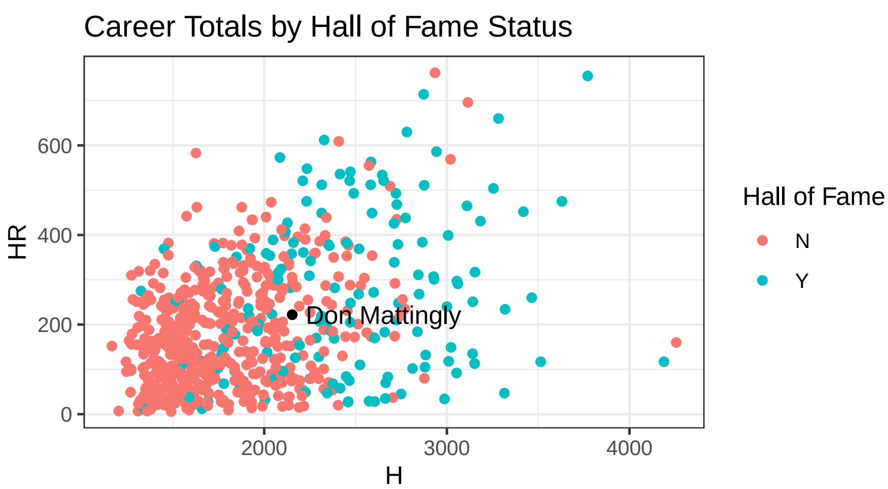
```
  geom_smooth(method = "lm",
              formula = y ~ x + I(x^2),
              se = FALSE) +
  labs(title = "Don Mattingly and Harold Baines")
```



How does he compare to other Hall of Famers?

```
library(glue)
library(plotly)
library(ggrepel)

vars <- c("G", "AB","R", "H", "X2B","X3B","HR", "RBI", "BB", "SO", "SB",
          "SH", "HBP", "SF")

# Career totals for players with 5000 at-bats
Batting %>%
  group_by(playerID) %>%
  summarize_at(vars, sum, na.rm = TRUE) %>%
  filter(AB > 5000) %>%
  mutate(AVG = round(H / AB, 3),
         HR.rate = HR / AB,
         PA = AB + BB + SH + HBP + SF) -> career_totals

# Must have retired at least five years ago
People %>%
```

```r
  filter(finalGame < 2018,
         finalGame > 1920) %>%
  pull(playerID) -> eligible
career_totals %>%
  filter(playerID %in% eligible) -> career_totals

# Determine whether they are in the hall of fame
career_totals %>%
  left_join(HallOfFame %>% filter(inducted == "Y", category == "Player") %>%
              select(playerID, yearID, inducted), by = "playerID") %>%
  replace_na(list(inducted = "N")) %>%
  rename(halloffame = inducted) -> career_totals

# Add full name
career_totals %>%
  left_join(People %>% select(nameLast, nameFirst, playerID)) %>%
  mutate(name = paste(nameFirst, nameLast, sep = " ")) %>%
  select(-nameLast, -nameFirst) -> career_totals

# Career Totals
p2 <- career_totals %>%
  ggplot(aes(label = name, x = H, y = HR, color = halloffame)) +
  geom_point() +
  geom_text(data = filter(career_totals, playerID == "mattido01"),
            aes(x = H + 500, y = HR, label = name), color = "black") +
  geom_point(data = filter(career_totals, playerID == "mattido01"),
             aes(x = H, y = HR), color = "black") +
  labs(title = "Career Totals by Hall of Fame Status") +
  theme_bw() +
  scale_color_discrete("Hall of Fame")
p2 # ggplotly(p2) # try ggplotly() to make this plot interactive
```



Predicting Hall of Fame Selection with Logistic Regression

```
# Here's a model based on aggregate statistics predicting Hall of Fame selection.
model.hof <- glm(halloffame == "Y" ~ H + X2B + X3B + HR + RBI + BB,
                 family = "binomial",
                 data = career_totals)
career_totals %>%
  mutate(hof_pred = predict(model.hof,
                            type = "response",
                            newdata = .)) -> career_totals
career_totals %>%
  arrange(desc(hof_pred)) %>%
  select(name, hof_pred, halloffame) %>%
  head(10) %>%
  kable(caption = "Top-10 Highest Probabilities for Hall of Fame Selection",
        col.names = c("Name", "P(HoF)", "Inducted"), digits = 3)
```

Table 2: Top-10 Highest Probabilities for Hall of Fame Selection

Name	P(HoF)	Inducted
Ty Cobb	1.000	Y
Hank Aaron	0.995	Y
Babe Ruth	0.994	Y

Name	P(HoF)	Inducted
Eddie Collins	0.993	Y
Stan Musial	0.993	Y
Willie Mays	0.986	Y
Lou Gehrig	0.986	Y
Tris Speaker	0.980	Y
Jimmie Foxx	0.975	Y
Al Simmons	0.974	Y

```
career_totals %>% arrange(desc(hof_pred)) %>%
  filter(hof_pred > 0.07, hof_pred < 0.075) %>%
  select(name, hof_pred, halloffame) %>%
  kable(caption = "Players with similar predictions as Don Mattingly",
        digits = 3)
```

Table 3: Players with similar predictions as Don Mattingly

name	hof_pred	halloffame
Jason Kendall	0.075	N
Adam Dunn	0.074	N
David Justice	0.074	N
Devon White	0.074	N
Harlond Clift	0.074	N
Mo Vaughn	0.073	N
Frank White	0.072	N
Benito Santiago	0.072	N
Dick Bartell	0.072	N
Fred Merkle	0.072	N
Hughie Critz	0.072	N
Frank Thomas	0.071	N
Reggie Sanders	0.071	N

Don Mattingly (not-selected) vs. Harold Baines (selected)

Let's look at Don Mattingly's and Harold Baines' career statistics side by side.

```
career_totals %>%
  filter(playerID %in% c("mattido01","baineha01")) %>%
  select(name, G, PA, AB, H, HR, AVG) %>%
  kable(caption = "Career Statistics", digits = 3)
```

Table 4: Career Statistics

name	G	PA	AB	H	HR	AVG
Harold Baines	2830	11092	9908	2866	384	0.289
Don Mattingly	1785	7721	7003	2153	222	0.307

Now, let's assume Mattingly and Baines both had 12,000 plate appearances.

Following the steps on page 47 of *Understanding Sabermetrics*, let's compare cumulative statistics for various kickers.

- Determine the projected number of new at-bats.
- Solve for x in $\frac{AB}{AB+BB} = \frac{AB+x}{newPA}$

This is the number of additional at-bats to award each hitter.

```
newPA = 12000

# Define "x" below as the number of additional at-bats.
career_totals %>%
  mutate(x = ((AB / (AB + BB)) * newPA - AB) %>% round(0)) -> career_totals

career_totals %>%
  filter(playerID %in% c("mattido01","baineha01")) %>%
  select(name, AB, x) %>%
  kable(caption = "Number of new at-bats.")
```

Table 5: Number of new at-bats.

name	AB	x
Harold Baines	9908	930
Don Mattingly	7003	4067

- Calculate the equivalence coefficient, $(1 + \frac{x}{AB} * k)$, where k is the kicker. Let's investigate $k \in \{1,\ 0.95,\ 0.90\}$.

```
# Function to compute equivalence coefficient
eqcoef <- function(k, df){
  df %>%
    mutate(k = k,
           EC = 1 + x / AB * k,
```

```
            H = round(H * EC, 0),
            X2B = round(X2B * EC, 0),
            X3B = round(X3B * EC, 0),
            HR = round(HR * EC, 0),
            RBI = round(RBI * EC, 0),
            BB = round(BB * EC, 0)) -> df
  return(df %>% select(playerID, name, k, EC, H, X2B, X3B, HR, RBI, BB))
}

k = c(1, 0.95, 0.90)
k %>%
  map_df(eqcoef, df = career_totals %>% filter(playerID == "mattido01")) -> matty_ec
k %>%
  map_df(eqcoef, df = career_totals %>% filter(playerID == "baineha01")) -> baine_ec
rbind(matty_ec, baine_ec) %>%
  kable(digits = 3)
```

playerID	name	k	EC	H	X2B	X3B	HR	RBI	BB
mattido01	Don Mattingly	1.00	1.581	3403	699	32	351	1737	929
mattido01	Don Mattingly	0.95	1.552	3341	686	31	344	1705	912
mattido01	Don Mattingly	0.90	1.523	3278	673	30	338	1673	895
baineha01	Harold Baines	1.00	1.094	3135	534	54	420	1781	1162
baineha01	Harold Baines	0.95	1.089	3122	532	53	418	1773	1157
baineha01	Harold Baines	0.90	1.084	3108	529	53	416	1766	1152

Don Mattingly Predictions with a Longer Career

```
matty_ec %>%
  mutate(hof_pred = predict(model.hof,
                            type = "response",
                            newdata = .)) -> matty_ec
matty_ec %>%
  select(name, k, hof_pred) %>%
  kable(digits = 3)
```

name	k	hof_pred
Don Mattingly	1.00	0.805
Don Mattingly	0.95	0.775
Don Mattingly	0.90	0.740

```
# Players similar to 'Hypothetical Mattingly'
career_totals %>%
  filter(hof_pred > 0.63, hof_pred < 0.72) %>%
  select(name, hof_pred, halloffame) %>%
  kable(digits = 3)
```

name	hof_pred	halloffame
Carlos Beltran	0.700	N
Lou Brock	0.653	Y
Max Carey	0.658	Y
Kiki Cuyler	0.677	Y
Jake Daubert	0.666	N
Willie Davis	0.646	N
Tony Gwynn	0.678	Y
Harry Hooper	0.645	Y
Chipper Jones	0.682	Y
Joe Judge	0.699	N
Harmon Killebrew	0.667	Y
Eddie Mathews	0.717	Y
Willie McCovey	0.663	Y
Stuffy McInnis	0.706	N
Joe Morgan	0.658	Y
Dave Parker	0.658	Y
Vada Pinson	0.640	N
Manny Ramirez	0.667	N
Brooks Robinson	0.718	Y
Mike Schmidt	0.671	Y
Sammy Sosa	0.689	N
Rusty Staub	0.714	N
Frank Thomas	0.633	Y
Jim Thome	0.632	Y
Bobby Veach	0.658	N